SESSION 18

# AIOps Governance, KPIs & Future Trends

- Measuring AIOps Success
- KPIs & Operational Metrics
- Future of AIOps & Autonomous Operations

AIOps Analytics

# Session 18 — Agenda

| | |
|---|---|
| 01 | **AIOps Governance Framework**<br>Principles, accountability & policy |
| 02 | **Measuring AIOps Success**<br>What good looks like & how to track it |
| 03 | **KPIs & Operational Metrics**<br>MTTR, alert noise, availability & more |
| 04 | **Building a Metrics Dashboard**<br>Instrumentation, tools & reporting |
| 05 | **AIOps Maturity Model**<br>Assessing & advancing your organisation |
| 06 | **Future Trends in AIOps**<br>AI, ML & next-gen autonomous operations |
| 07 | **Autonomous Operations**<br>Self-healing, self-optimising systems |

# What is AIOps Governance?

*AIOps Governance is the framework of policies, processes, and accountability structures that ensure AI-driven IT operations deliver trusted, measurable, and compliant outcomes.*

### Accountability

Clear ownership of AI decisions & outcomes

### Transparency

Explainable models and audit trails

### Compliance

Adherence to SLAs, regulations & policies

### Continuous Eval

Ongoing monitoring of model drift & accuracy

# AIOps Governance Pillars

### Data Governance

Data quality, lineage, catalog & stewardship

### Model Governance

Versioning, drift detection, explainability (XAI)

### Process Governance

Change management, runbooks & approval gates

### People & Culture

Roles, training, AI ethics & oversight

### Infrastructure Gov.

Cloud policy, cost controls, security compliance

### Outcome Governance

KPI alignment, SLAs, value realisation

# Measuring AIOps Success

*"You cannot manage what you cannot measure. AIOps outcomes must be observable, quantifiable, and aligned to business value."*

| Operational | Business | AI Model | Adoption |
| --- | --- | --- | --- |
| Speed, reliability, incident reduction | Uptime, revenue impact, customer satisfaction | Accuracy, false-positive rate, coverage | Team usage, automation ratio, tool penetration |

Source: Forrester – Measuring AIOps Value 2024 | https://www.forrester.com/report/measuring-aiops-value

# Before vs After AIOps — Success Signals

| Dimension | Before AIOps | After AIOps |
|---|---|---|
| Alert Noise | 500–1000+ alerts/day | 80–90% noise reduction |
| MTTR | Hours to resolve | Minutes (automated RCA) |
| Incident Volume | Reactive firefighting | Proactive prevention |
| On-call Burden | High engineer fatigue | Reduced pager-duty load |
| Root Cause | Manual log analysis (hours) | AI-identified in <5 min |
| Capacity Planning | Monthly manual reviews | Continuous auto-scaling |
| SLA Compliance | Reactive breach response | Predictive SLA assurance |

Source: PagerDuty State of Digital Ops 2024 | https://www.pagerduty.com/resources/reports/state-of-digital-operations/

# Core AIOps KPIs at a Glance

**MTTR**
Mean Time to Resolve
**Target: ↓ 70%**

**MTTD**
Mean Time to Detect
**Target: <2 min**

**Alert Noise**
False Positive Rate
**Target: <5%**

**Availability**
Service Uptime
**Target: 99.99%**

**Automation**
% Incidents Auto-resolved
**Target: >60%**

**Model Acc.**
AIOps ML Accuracy
**Target: >95%**

KPI Metrics

# KPI: Service Availability & SLA Compliance

| Availability | Annual Downtime | Monthly Downtime | Required Ops Level |
|---|---|---|---|
| 99% | ~87.6 hrs/year | ~7.3 hrs/mo | Traditional monitoring |
| 99.9% | ~8.76 hrs/year | ~43.8 min/mo | Basic AIOps |
| 99.95% | ~4.38 hrs/year | ~21.9 min/mo | AIOps + automation |
| 99.99% | ~52.56 min/year | ~4.38 min/mo | Full AIOps suite |
| 99.999% | ~5.26 min/year | ~25.9 sec/mo | Autonomous operations |

### Availability SLA

Uptime guarantees (99.x%)

### Performance SLA

Response time, latency thresholds

### Recovery SLA

RTO & RPO targets

# KPI: Incident Reduction & Automation Rate

### P1/P2 Incident Volume

# critical incidents per week

**↓ 40–60% within 6 months**

### Automation Rate

% incidents resolved w/o human

**Target: >60%**

### Runbook Execution

Automated vs manual remediation

**80:20 auto:manual ratio**

### Repeat Incidents

Recurrence of known issues

**↓ 70% via ML pattern match**

IT Operations

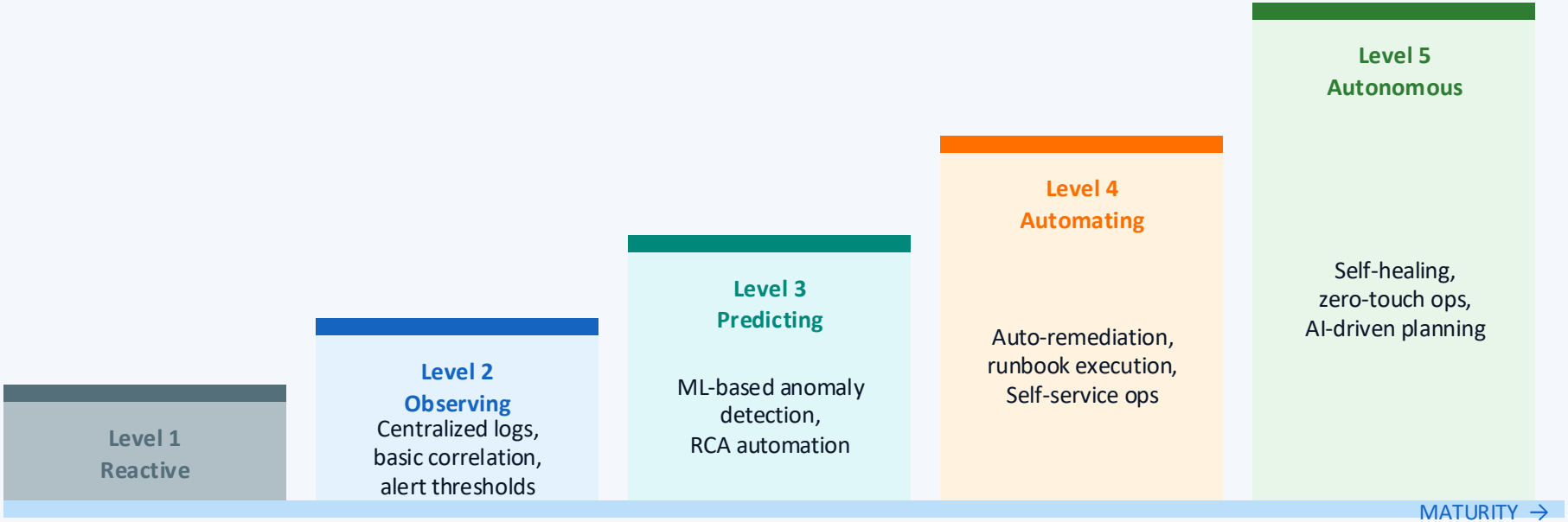
# Building an AIOps Metrics Dashboard

| | |
|---|---|
| **Executive Layer** | C-suite OKRs: Availability, Cost, Customer Impact |
| **Operational Layer** | MTTR, MTTD, Alert Volume, Automation Rate |
| **AI/ML Layer** | Model accuracy, drift alerts, confidence scores |
| **Infrastructure Layer** | CPU, memory, network, latency, saturation |

**Popular Tooling:**

- Grafana
- Datadog
- Dynatrace
- Splunk
- New Relic
- ServiceNow

# AIOps Maturity Model



# The Future of AIOps

*AIOps is evolving from reactive monitoring to fully autonomous, self-optimising systems powered by Generative AI, advanced ML, and cross-domain intelligence.*

## GenAI + LLMs

Natural language queries,
AI-authored runbooks

## Causal AI

True root cause, not
just correlations

## Edge & 5G AIOps

Ultra-low latency ops
at the edge

## FinOps Integration

Cost-aware intelligent
resource optimisation

## Security AIOps

Unified AIOps + SecOps
threat correlation

## Sustainability

Green ops: AI-driven
energy optimisation

# Generative AI & LLMs Transforming AIOps

## Use Cases

### NL Incident Query

Ask 'What caused last week's outage?' in plain English

### Auto-Runbook Gen.

LLM drafts remediation steps from incident history

### Log Summarisation

Condense 10K log lines into 5-sentence RCA

### Postmortem Drafting

AI-authored incident retrospectives

### Chatbot Ops Assistant

Conversational interface for on-call engineers

## GenAI AIOps Architecture

Observability Data
(Logs, Metrics, Traces)

↓

Embedding & Vector Store

↓

LLM (GPT-4 / Llama 3 / Gemini)

↓

Retrieval-Augmented Generation

↓

Incident Response Action

# Autonomous IT Operations

## The OODA Loop for Autonomous Ops

| | |
|---|---|
| **OBSERVE** | Continuous telemetry collection:<br>metrics, logs, traces, events |
| **ORIENT** | AI contextualisation:<br>anomaly detection + correlation |
| **DECIDE** | Causal reasoning:<br>recommend or auto-select action |
| **ACT** | Automated execution:<br>self-healing, auto-scaling, runbooks |

Autonomous Ops

# Self-Healing Systems Architecture

**Anomaly Detection**

**Predictive Alerts**

**Auto-Remediation**

## Self-Healing Engine

**Root Cause Analysis**

- ↓ 80% manual toil
- 99.99%+ uptime
- Auto-scales proactively
- 24/7 zero-touch ops
- Continuous self-improvement

**Capacity Auto-Scale**

Source: Red Hat – Self-Healing with AIOps | https://www.redhat.com/en/blog/self-healing-it-operations

# AIOps Roadmap: 2025–2030

## 2025
Foundation

- Unified observability
- Alert deduplication
- Basic ML models

## 2026
Intelligence

- GenAI-powered RCA
- LLM runbooks
- Predictive incident mgmt

## 2027
Automation

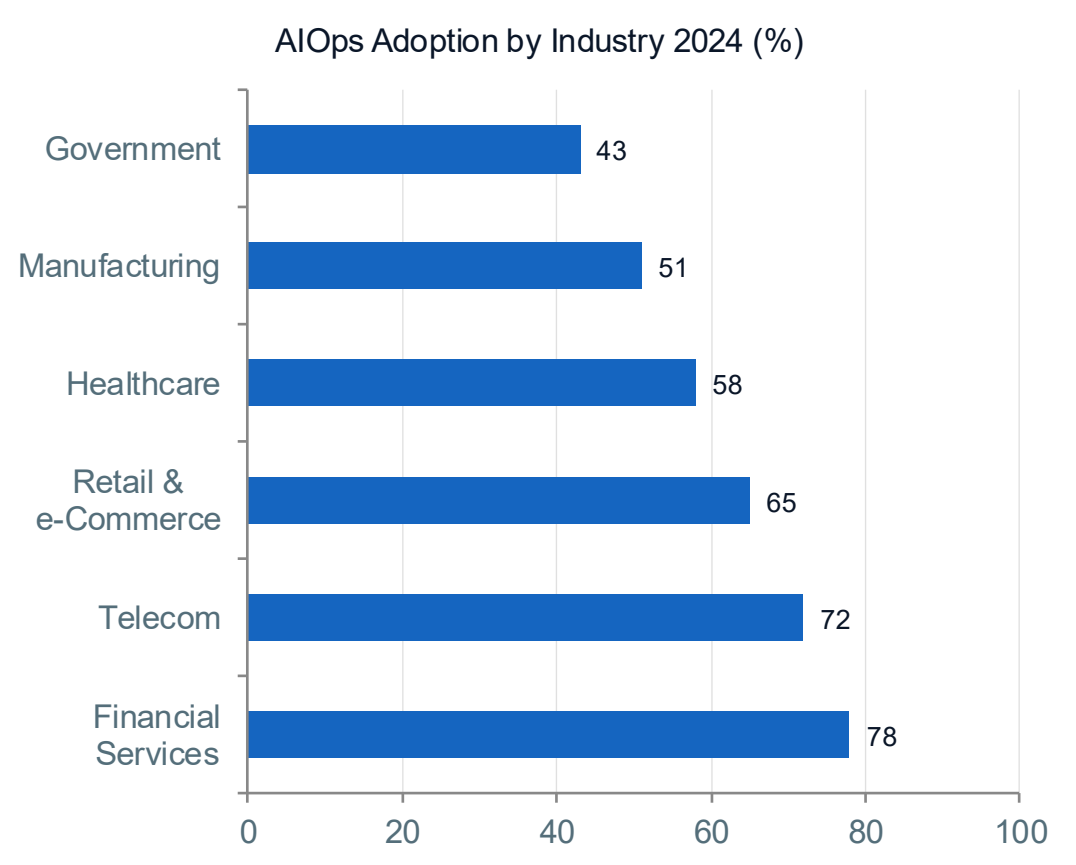
- Full auto-remediation
- ChatOps integration
- AIOps + FinOps

## 2028–30
Autonomy

- Zero-touch operations
- Self-optimising infra
- AI SRE agents

Source: Forrester AIOps Forecast 2030 | https://www.forrester.com/report/the-aiops-roadmap-2030/

# AIOps Market Adoption & Industry Trends




**$40.9B**

Global AIOps market size by 2028

**37.9%**

CAGR 2023–2028

**68%**

Enterprises using AIOps for at least one domain

**3.2x**

ROI reported by mature AIOps adopters

# Challenges & Risks in AIOps Governance

### Model Drift
**HIGH**

ML models degrade over time as IT environments change

### Alert Fatigue 2.0
**HIGH**

AI-generated alerts can overwhelm teams if tuned poorly

### Explainability Gap
**MEDIUM**

Black-box AI decisions erode operator trust

### Data Quality Issues
**HIGH**

Poor telemetry data → garbage in, garbage out

### Skills Gap
**MEDIUM**

Shortage of ML-literate SREs and IT ops engineers

### Vendor Lock-in
**MEDIUM**

Dependency on proprietary AIOps platforms

### Security & Privacy
**HIGH**

AI models trained on sensitive operational data

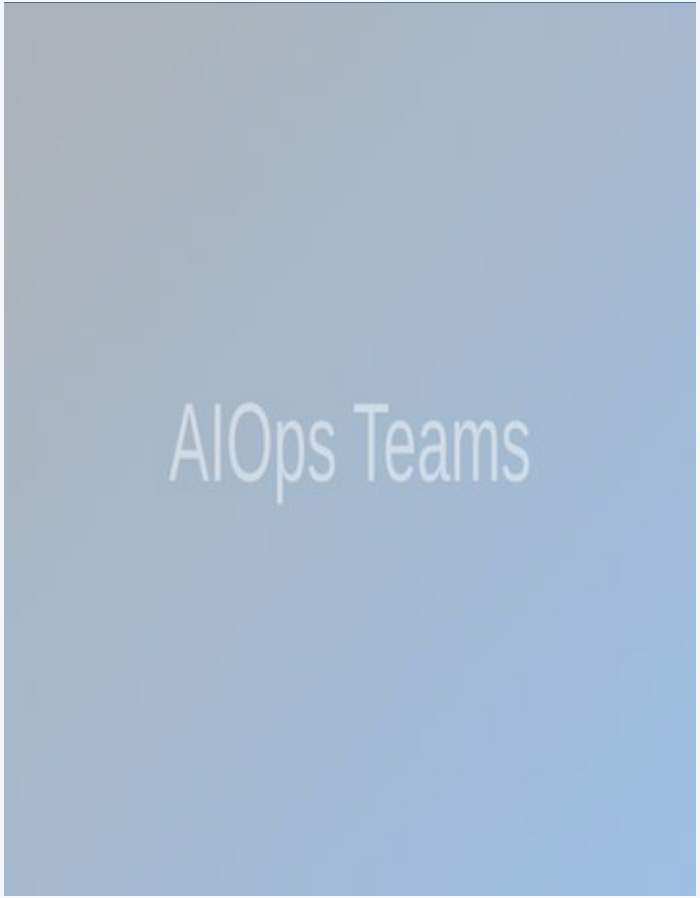
### Over-automation Risk
**HIGH**

Automated actions without human review can cascade failures

Source: ISACA – AI Risk in IT Operations | https://www.isaca.org/resources/news-and-trends/newsletters/atisaca/2024/ai-ops-risk

# Best Practices for AIOps Governance

1. Define clear success metrics before deployment — agree KPIs with business stakeholders
2. Implement model monitoring: track drift, accuracy, and coverage continuously
3. Start with augmentation, not automation — build trust incrementally
4. Maintain a human-in-the-loop for P1 automated actions initially
5. Create an AIOps Centre of Excellence (CoE) with cross-functional ownership
6. Version-control all ML models and maintain full audit logs
7. Align AIOps governance with ITIL 4, COBIT, and ISO/IEC 27001




Source: ITIL 4 AIOps Integration Guide | https://www.axelos.com/certifications/itil-service-management

# Session 18 — Key Takeaways

## Governance is Critical

AIOps without governance leads to ungoverned AI, model drift, and eroded trust. Establish clear policies and ownership.

## KPIs Drive Value

Measure MTTR, MTTD, alert noise, automation rate, and availability. Tie every metric back to business outcomes.

## Maturity is a Journey

Start with visibility, build toward intelligence, and progressively automate — don't rush to autonomy.

## GenAI is a Game Changer

LLMs and generative AI are transforming incident management, RCA, and runbook authoring at scale.

## Autonomous Ops is the Goal

The future is self-healing, zero-touch systems — but humans remain essential for oversight and complex decisions.